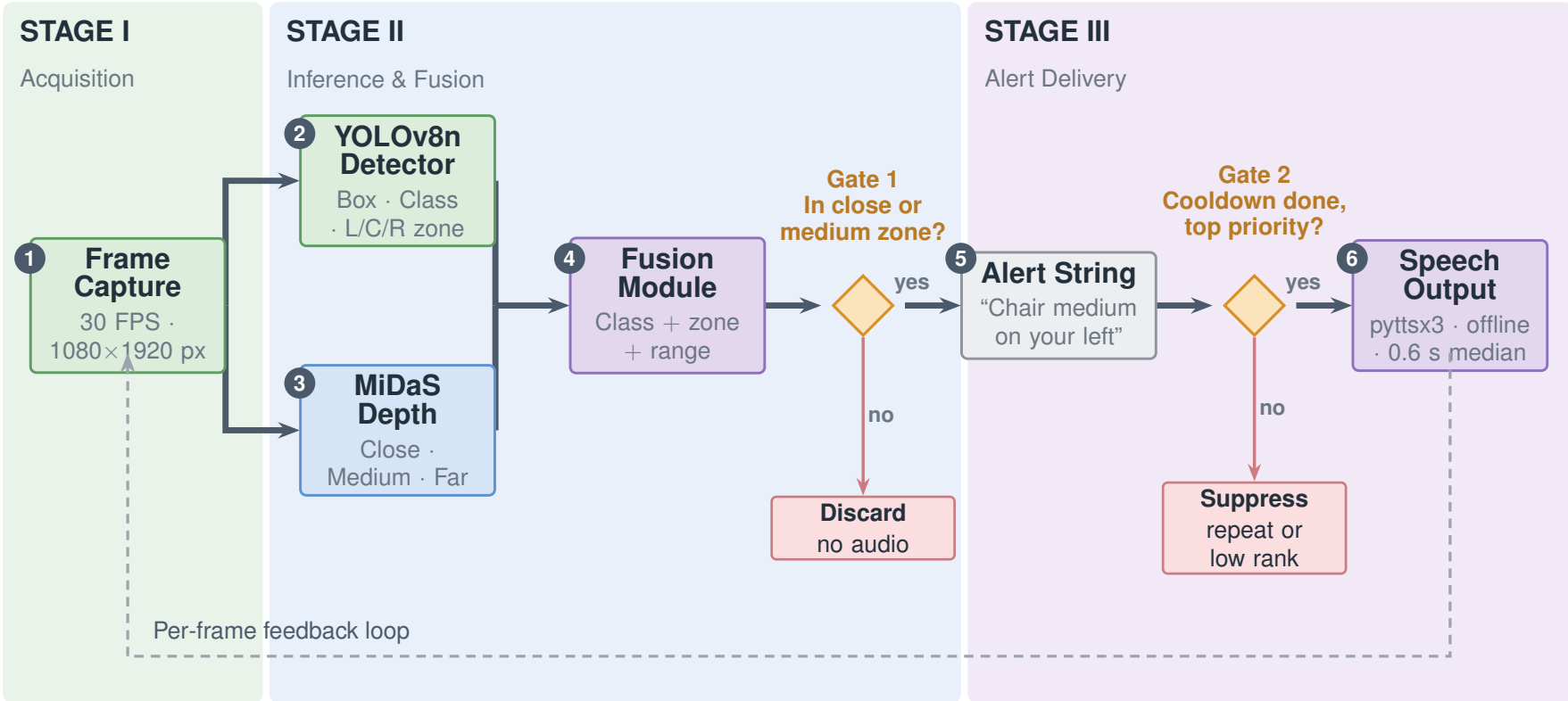
STAGE I
Acquisition
STAGE II
Inference & Fusion
STAGE III
Alert Delivery
1 Frame Capture
30 FPS · 1080×1920 px
2 YOLOv8n Detector
Box · Class · L/C/R zone
3 MiDaS Depth
Close · Medium · Far
4 Fusion Module
Class + zone + range
Gate 1
In close or medium zone?
yes
no
Discard
no audio
5 Alert String
“Chair medium on your left”
Gate 2
Cooldown done, top priority?
yes
no
Suppress
repeat or low rank
6 Speech Output
pyttsx3 · offline · 0.6 s median
Per-frame feedback loop